

Lecture Scribe Exam Reference



Tutorial_2.pdf

PDF



Tutorial_2_Solution.pdf

PDF

Create a lecture scribe to be used as exam oriented reference content to be used with a reading based / closed notes examination.

Work with the attached lecture slides/ PDF and the corresponding textbook materials as the background information only. Do not use outside knowledge. Do not make any inferences not directly available in the given documents.

The tutorial session in the lecture was problem solving. Restructure the session accurately and fully.

The scribe must:

- Adhere to the arrangement and sequence of the subjects as in the lecture materials.
- Provide all definitions and notation.
- Indicate all assumptions and conditions.
- Give all statement of the theorems or formal results which occur in the set-up.
- Current evidence and inferences step by step just as presented on the materials.
- Write out all of the examples and solved problems in stages.
- Maintain rational relations among ideas and solutions.
- Be precise mathematically.

-Write clear headings of sections and systematic arrangement befitting revision.

The scribe must not:

-Introduce intuition, comments, simplifications, or imaginative explanations that are not already found in the materials.

-Present new examples or other approaches to solutions.

-Eliminate the middle reasoning steps that are discussed in the lecture.

-Rearrange the material so as to alter the logical sequence of the lecture.

-Justify your rationale.

The finished work should be a faithful copy of what was being taught to the point that a student will only use this paper to refresh the lecture materials to take an exam.

CSE 400: Fundamentals of Probability in Computing

Tutorial-2 Lecture Scribe (Exam-Oriented Reference)

School of Engineering and Applied Science (SEAS), Ahmedabad University

Tutorial Session: Problem Solving

Based strictly on Tutorial-2 and Tutorial-2: Solution

Q1. Random Point on a Line Segment

Problem Statement

A point is chosen at random on a line segment of length L . Interpret this statement, and find the probability that the ratio of the shorter to the longer segment is less than $\frac{1}{4}$.

Interpretation

Let the point be chosen at a distance x from one end, where:

$$0 < x < L$$

Two segments are formed:

- Length x
- Length $L - x$

The ratio of the shorter to the longer segment is:

$$R = \frac{\min(x, L - x)}{\max(x, L - x)}$$

We compute:

$$P(R < \frac{1}{4})$$

Case Analysis (Based on Midpoint)

Case 1: $x \leq \frac{L}{2}$

Shorter segment: x

Longer segment: $L - x$

$$\frac{x}{L - x} < \frac{1}{4}$$

$$4x < L - x$$

$$5x < L$$

$$x < \frac{L}{5}$$

Case 2: $x > \frac{L}{2}$

Shorter segment: $L - x$

Longer segment: x

$$\frac{L-x}{x} < \frac{1}{4}$$

$$4(L-x) < x$$

$$4L - 4x < x$$

$$5x > 4L$$

$$x > \frac{4L}{5}$$

Valid Region

$$x \in (0, \frac{L}{5}) \cup (\frac{4L}{5}, L)$$

Total valid length:

$$(\frac{L}{5} - 0) + (L - \frac{4L}{5}) = \frac{L}{5} + \frac{L}{5} = \frac{2L}{5}$$

Since the point is uniformly chosen:

$$P = \frac{2L/5}{L} = \frac{2}{5}$$

Q2. Bayesian Classification with Normal Distributions

Given

White region:

$$X | W \sim N(4, 4), \quad \sigma_1 = 2$$

Black region:

$$X | B \sim N(6, 9), \quad \sigma_2 = 3$$

$$P(B) = \alpha, \quad P(W) = 1 - \alpha$$

Observed value:

$$X = 5$$

Condition:

$$P(B \mid X = 5) = P(W \mid X = 5) = \frac{1}{2}$$

Bayes' Theorem

$$P(B \mid X = 5) = \frac{\alpha f(5 \mid B)}{\alpha f(5 \mid B) + (1 - \alpha) f(5 \mid W)}$$

Set equal to $\frac{1}{2}$.

Normal Density Formula

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Compute:

$$f(5 \mid B) = \frac{1}{3\sqrt{2\pi}} e^{-1/18}$$

$$f(5 \mid W) = \frac{1}{2\sqrt{2\pi}} e^{-1/8}$$

Substitute:

$$\frac{\alpha e^{-1/18}/(3\sqrt{2\pi})}{\alpha e^{-1/18}/(3\sqrt{2\pi}) + (1 - \alpha) e^{-1/8}/(2\sqrt{2\pi})} = \frac{1}{2}$$

Simplifies to:

$$\frac{3\alpha}{3\alpha + 2(1 - \alpha)e^{-5/72}} = \frac{1}{2}$$

Using $e^{-5/72} \approx 0.93$:

$$6\alpha = 3\alpha + 2(1 - \alpha)(0.93)$$

$$3\alpha = 1.86 - 1.86\alpha$$

$$4.86\alpha = 1.86$$

$$\alpha \approx 0.3827$$

Q3. Minimizing Expected Distance

(a) Finite Road (0, A)

$$X \sim \text{Uniform}(0, A)$$

$$f_X(x) = \frac{1}{A}$$

Minimize:

$$E[|X - a|] = \int_0^a (a - x) \frac{1}{A} dx + \int_a^A (x - a) \frac{1}{A} dx$$

After evaluation:

$$\frac{2a^2 - 2aA + A^2}{2A}$$

Differentiate:

$$\frac{1}{2A}(4a - 2A) = 0$$

$$4a = 2A$$

$$a = \frac{A}{2}$$

(b) Infinite Road, Exponential Distribution

$$X \sim \text{Exponential}(\lambda)$$

$$f_X(x) = \lambda e^{-\lambda x}$$

Minimize:

$$E[|X - a|] = \int_0^a (a - x) \lambda e^{-\lambda x} dx + \int_a^\infty (x - a) \lambda e^{-\lambda x} dx$$

Derivative:

$$\frac{d}{da} E[|X - a|] = 1 - 2e^{-\lambda a}$$

Set equal to 0:

$$2e^{-\lambda a} = 1$$

$$e^{-\lambda a} = \frac{1}{2}$$

$$a = \frac{\ln 2}{\lambda}$$

Q4. Mixture of Exponentials

Given

Type i battery lifetime:

$$X \mid T = i \sim \text{Exponential}(\lambda_i)$$

$$P(T = i) = p_i, \quad p_1 + p_2 = 1$$

Find:

$$P(X > t + s \mid X > t)$$

Law of Total Probability

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

$$P(X > t + s) = p_1 e^{-\lambda_1(t+s)} + p_2 e^{-\lambda_2(t+s)}$$

Conditional probability:

$$\frac{p_1 e^{-\lambda_1(t+s)} + p_2 e^{-\lambda_2(t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}$$

Q5. Conditional Distribution of Poisson Variables

Given

$$X \sim \text{Poisson}(\lambda_1)$$

$$Y \sim \text{Poisson}(\lambda_2)$$

Independent.

Find:

$$P(X = k \mid X + Y = n)$$

$$P(X = k \mid X + Y = n) = \frac{P(X = k)P(Y = n - k)}{P(X + Y = n)}$$

After substitution and simplification:

$$= \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{n-k}$$

Q6. Transformation of Random Variables

Given

$$X \sim \text{Uniform}[-2, 2]$$

$$f_X(x) = \frac{1}{4}$$

$$g(x) = \begin{cases} x, & x \in [-2, -1] \\ 0, & x \in (-1, 1) \\ x, & x \in [1, 2] \end{cases}$$

$$Y = g(X)$$

CDF $F_Y(y)$

Piecewise:

$$F_Y(y) = 0 \quad y < -2$$

$$F_Y(y) = \frac{y+2}{4}, \quad -2 \leq y < -1$$

$$F_Y(y) = \frac{1}{4}, \quad -1 \leq y < 0$$

At $y = 0$:

$$P(Y = 0) = \frac{1}{2}$$

$$F_Y(0) = \frac{3}{4}$$

$$F_Y(y) = \frac{3}{4}, \quad 0 < y < 1$$

$$F_Y(y) = \frac{y+2}{4}, \quad 1 \leq y \leq 2$$

$$F_Y(y) = 1, \quad y > 2$$

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$$f_Y(y) = \frac{1}{4}, \quad -2 < y < -1$$

$$f_Y(y) = \frac{1}{4}, \quad 1 < y < 2$$

Point mass:

$$P(Y = 0) = \frac{1}{2}$$

Q7. Binomial Stock Model

Given

$$u = 1.012, \quad d = 0.990, \quad p = 0.52$$

After 1000 periods:

$$S_{1000} = S_0 u^k d^{1000-k}$$

Condition:

$$S_{1000} \geq 1.30 S_0$$

Taking logs:

$$k(\ln u - \ln d) + 1000 \ln d \geq \ln(1.30)$$

Numerically:

$$k \geq 469.4$$

$$k \geq 470$$

$$k \sim \text{Binomial}(1000, 0.52)$$

Required probability:

$$\sum_{k=470}^{1000} \binom{1000}{k} (0.52)^k (0.48)^{1000-k}$$

Q8. Exponential Repair Time

$$X \sim \text{Exponential}\left(\frac{1}{2}\right)$$

$$f(x) = \frac{1}{2} e^{-x/2}$$

(a)

$$P(X > 2) = \int_2^{\infty} \frac{1}{2} e^{-x/2} dx = e^{-1}$$

(b)

$$P(X \geq 10 \mid X > 9) = \frac{e^{-5}}{e^{-9/2}} = e^{-1/2}$$

Q9. Distribution of $Z = \sqrt{X^2 + Y^2}$

$$F_Z(z) = P(X^2 + Y^2 \leq z^2)$$

$$= \int_{-z}^z \int_{-\sqrt{z^2-x^2}}^{\sqrt{z^2-x^2}} f_X(x) f_Y(y) dy dx$$

Differentiating:

$$f_Z(z) = z \int_{-z}^z \frac{f_X(x) [f_Y(\sqrt{z^2-x^2}) + f_Y(-\sqrt{z^2-x^2})]}{\sqrt{z^2-x^2}} dx$$

Q10. Joint Distribution of M_n and M_{n+1}

$$M_n = \max(X_1, \dots, X_n)$$

$$M_{n+1} = \max(M_n, X_{n+1})$$

$$P(M_n \leq x) = F(x)^n$$

Joint CDF

Case 1: $x \leq y$

$$P(M_n \leq x, M_{n+1} \leq y) = F(x)^n F(y)$$

Case 2: $x > y$

$$P(M_n \leq x, M_{n+1} \leq y) = F(y)^{n+1}$$

Final Result

$$P(M_n \leq x, M_{n+1} \leq y) = \begin{cases} F(x)^n F(y), & x \leq y \\ F(y)^{n+1}, & x > y \end{cases}$$

End of Tutorial-2 Lecture Scribe

(Complete, sequential, exam-oriented reproduction of lecture material.)